

Reasoning analysis — gpt-5.6-sol (50 runs, `real_gpt56sol_med_50`)

Each model rationale is split into sentences; sentences making the same point across runs are grouped. **Shared** = the point appears in at least 60% of the runs in that group; **varying** = it appears in some runs but not most. Counts are runs, not sentences. 'Evidence cited' lists the numbers and times the model quoted, with the number of runs quoting each.

decision: Profile / basal 00:00

keep — 50/50 runs

Shared reasoning

- (30/50) Low exposure is elevated from midnight to 03:00, but residual evening IOB, the 90 mg/dL target, SMB/UAM, Dynamic ISF, and corrections remain unresolved.

Varying reasoning

- (9/50) Keep basal while testing the 00:00 target alone; reassessment depends on fasting-like nights with complete IOB context.
- (8/50) Fasting-like overnight observations are required before attributing the pattern to this basal segment.
- (6/50) Evidence shows recurrent lows from local 00:00-03:00, but uncertainty remains about preceding corrections, SMB/UAM, activity, and low treatment.
- (4/50) Keep pending fasting-like, low-IOB overnight verification.
- (4/50) Keep this segment until low-IOB overnight observations separate scheduled basal from those dependencies.
- (3/50) Evidence shows overnight lows, but uncertainty remains because prior corrections, SMB/UAM, Dynamic ISF, activity, and the 90 mg/dL target may contribute.
- ... 12 more varying points

Points made by a single run only: 15.

Phrases used by at least half the runs: "evening iob" (32), "90 mg/dl" (28), "90 mg/dl target" (27).

Evidence cited: 90mg/dl (28), 00:00 (17), 03:00 (13), 23:00 (3), 0.6u/h (3), 04:00 (2), 08:00 (1), 19:00 (1).

decision: Profile / basal 05:00

keep — 50/50 runs

Varying reasoning

- (17/50) Local 05:00-06:00 glucose is comparatively stable with 95.2-97.5% TIR.
- (5/50) One isolated severe reading does not isolate this segment, and sensor reliability, IOB, and activity remain uncertain.
- (5/50) Local glucose is generally stable around 05:00–06:00, although some low exposure remains.
- (5/50) The 05:00 hour averaged 104.2 mg/dL with 4.8% below range, but no fasting-labelled data isolate basal.
- (4/50) Evidence shows comparatively stable glucose from 04:00-08:00.
- (4/50) Meal, activity, and IOB uncertainty prevents a supported change.
- ... 16 more varying points

Points made by a single run only: 27.

Evidence cited: 05:00 (32), 06:00 (21), 04:00 (13), 4.8% (6), 104.2mg/dl (6), 08:00 (5), 95.2% (5), 97.5% (4), 99.5mg/dl (3), 00:00 (3), 03:00 (3), 07:00 (2).

decision: Profile / basal 06:00

keep — 50/50 runs

Shared reasoning

- (37/50) Local 06:00–08:00 has 97.5–98.7% time in range and little low exposure.

Varying reasoning

- (5/50) The later rise could be meal-related, so neither an increase nor decrease is supported without fasting and meal-linked evidence.
- (4/50) The 06:00 to 08:00 period is comparatively stable, with means near 100 to 111 mg/dL and low TBR.
- (4/50) One isolated 42 mg/dL event and missing insulin and activity context do not support changing this segment.
- (3/50) Below-range exposure is low around 07:00–08:00, while the later rise could reflect breakfast or dawn effects.
- (3/50) The isolated 42 mg/dL minimum does not establish a recurring segment defect, and prior IOB or sensor effects remain uncertain.
- (2/50) The isolated 42 mg/dL morning reading and absent meal/activity data create uncertainty; meal-free observation is required before changing this segment.
- ... 8 more varying points

Points made by a single run only: 30.

Evidence cited: 06:00 (44), 08:00 (38), 42mg/dl (16), 07:00 (10), 98.7% (9), 97.5% (6), 07:20 (4), 111mg/dl (2), 05:00 (1), 07:59 (1), 2.5% (1), 98% (1).

decision: Profile / basal 10:00

keep — 50/50 runs

Varying reasoning

- (9/50) Late-morning and afternoon highs may reflect unrecorded meals, bolus timing, CR, UAM, or Dynamic ISF rather than basal.
- (5/50) Daytime highs are present, but absent meal records, UAM dosing, Dynamic ISF, corrections, temporary targets, and delayed absorption prevent attribution to basal.
- (5/50) Subsequent evening lows make an increase unsafe to infer.
- (4/50) Evidence shows daytime highs, but they occur in meal-like hours with no carbohydrate records.
- (3/50) Daytime glucose rises are observed, but zero carbohydrate records and COB-0 UAM responses make meals, bolus timing, and automation more plausible than proven basal deficiency.
- (3/50) This segment depends on meal timing, CR, SMB/UAM, Dynamic ISF, and IOB, so a basal change is unsupported.
- ... 12 more varying points

Points made by a single run only: 45.

Phrases used by at least half the runs: “dynamic isf” (29).

Evidence cited: 10:00 (2), 18:00 (1), 19:00 (1), 08:00 (1), 09:00 (1).

decision: Profile / basal 16:00

keep — 50/50 runs

Shared reasoning

- (32/50) High exposure near 16:00 is followed by increased low exposure at 18:00–19:00.

Varying reasoning

- (4/50) Meal insulin, Dynamic ISF, SMB/UAM, corrections, and IOB are unresolved dependencies, so increasing this segment is unsupported.
- (4/50) Meal and automation effects must be separated before changing basal.
- (3/50) UAM, SMB, Dynamic ISF, and IOB provide stronger competing explanations than insufficient basal.
- (3/50) Meal absorption and accumulated automated insulin must be separated before attributing the rise to basal.
- (3/50) Increasing basal could worsen delayed lows, while decreasing it is unsupported without fasting-like data; meal insulin and IOB remain major dependencies.
- (2/50) The 16:00 hour has the highest mean glucose and above-range exposure, but it is followed by increased TBR at 18:00–19:00.
- ... 10 more varying points

Points made by a single run only: 33.

Phrases used by at least half the runs: “low exposure” (27).

Evidence cited: 16:00 (42), 18:00 (27), 19:00 (27), 17:00 (11), 0.5u/h (1).

decision: Profile / basal 18:00

keep — 50/50 runs

Varying reasoning

- (13/50) Time below range is 7.7% at 18:00, but preceding daytime corrections, SMB/UAM, Dynamic ISF, activity, and rescue treatment are unresolved.
- (11/50) Evidence shows elevated low exposure around 18:00-19:00, but preceding automated and correction insulin is a strong competing cause.
- (4/50) Complete IOB and meal context are required before attributing the pattern to this basal segment.
- (3/50) Severe lows occur near this period, but preceding afternoon corrections, SMB/UAM, Dynamic ISF, activity, and meal insulin are unresolved dependencies.
- (3/50) Low exposure rises during this segment, but preceding SMB, temporary basal, corrections, meal effects, and activity are unresolved.
- (3/50) A basal change is unsupported until those dependencies are reconstructed.
- ... 12 more varying points

Points made by a single run only: 41.

Evidence cited: 18:00 (30), 7.7% (11), 19:00 (9), 16:00 (2), 15.3% (1).

decision: Profile / basal 19:00

keep — 50/50 runs

Varying reasoning

- (8/50) Time below range reaches 15.3% at 19:00, creating a safety concern, but the pattern follows afternoon highs and substantial reactive insulin.
- (7/50) Local TBR reached 15.3% at 19:00, creating concern, but afternoon SMB, UAM, temporary basal, corrections, activity, and low treatment are unresolved.
- (6/50) Local TBR is 15.3% at 19:00, but this follows afternoon highs and substantial automated insulin exposure.
- (5/50) Time below range is 15.3% at 19:00, but severe events follow periods of corrections, SMB/UAM, Dynamic ISF, and temporary basal.
- (4/50) The 19:00 hour has 15.3% below range, but accumulated daytime insulin is a strong alternative explanation.
- (4/50) Uncertainty prevents a specific basal reduction; review depends on complete preceding IOB and activity data.

- ... 10 more varying points

Points made by a single run only: 46.

Phrases used by at least half the runs: “below range” (25).

Evidence cited: 19:00 (42), 15.3% (38), 23:00 (2), 22:00 (1), 01:00 (1), 00:00 (1).

decision: Profile / CR 00:00

keep — 50/50 runs

Varying reasoning

- (14/50) No carbohydrate entries or meal-linked responses exist for this segment.
- (10/50) Overnight lows cannot be assigned to CR rather than basal, target, corrections, or IOB.
- (8/50) Overnight lows may involve residual meal insulin, but CR cannot be separated from basal, target, Dynamic ISF, SMB/UAM, and IOB.
- (8/50) There are no meal-linked carbohydrate or bolus records for this segment.
- (6/50) No carbohydrate amounts or identifiable meal boluses are available for this segment.
- (5/50) No meal carbohydrate amount or linked meal bolus is available for this segment.
- ... 12 more varying points

Points made by a single run only: 26.

Phrases used by at least half the runs: “meal linked” (32), “overnight lows” (31).

decision: Profile / CR 04:00

keep — 50/50 runs

Varying reasoning

- (7/50) Morning glucose rises are not linked to known meals, carbohydrate amounts, prebolus timing, or immediate bolus percentage.
- (5/50) The morning rise may reflect meal timing or unannounced carbohydrates, but carb amount and prebolus execution are absent.
- (5/50) Morning rises could reflect meal timing, unannounced carbohydrates, UAM response, or absorption.
- (4/50) The early-morning period is comparatively stable, but no carbohydrate amount or meal bolus is recorded.
- (4/50) CR assessment depends on a documented meal and six-hour response.
- (4/50) The comparatively stable early-morning glucose pattern does not independently support a CR change.
- ... 16 more varying points

Points made by a single run only: 36.

Evidence cited: 06:00 (1), 08:00 (1), 04:00 (1).

decision: Profile / CR 08:00

keep — 50/50 runs

Varying reasoning

- (7/50) Morning and afternoon rises may reflect unannounced meals, timing, or absorption.
- (6/50) Daytime rises could reflect unannounced meals, timing, absorption, or UAM response rather than a weak CR.
- (5/50) Later lows and accumulated IOB make a stronger CR unsafe to infer without meal-linked evidence.
- (3/50) Morning and afternoon rises could suggest insufficient meal coverage, but no carbohydrates, meal boluses, or prebolus data are available.
- (3/50) Daytime highs could suggest insufficient or delayed meal insulin, but no carbohydrate amount, prebolus interval, or immediate percentage exists.
- (3/50) Later low risk argues against strengthening it without a matched six-hour meal record.
- ... 13 more varying points

Points made by a single run only: 46.

Evidence cited: 10:00 (3), 08:00 (2), 17:00 (1).

decision: Profile / CR 16:00

keep — 50/50 runs

Varying reasoning

- (9/50) Afternoon highs are followed by evening lows.
- (6/50) Evidence shows afternoon highs followed by substantial 18:00–19:00 low exposure.
- (6/50) Dynamic ISF, SMB/UAM, corrections, and IOB must be separated first.
- (5/50) This stronger CR overlaps the daytime high-to-evening-low transition.
- (4/50) Evidence: this is the strongest CR segment and overlaps the afternoon high-to-evening-low pattern.
- (4/50) This is already the strongest CR segment, and afternoon highs are followed by increased evening low exposure.
- ... 13 more varying points

Points made by a single run only: 36.

Phrases used by at least half the runs: “afternoon highs” (26).

Evidence cited: 18:00 (3), 19:00 (3), 16:00 (2).

decision: Profile / CR 20:00

keep — 50/50 runs

Varying reasoning

- (15/50) Scheduled basal, target, corrections, automation, and residual IOB remain unresolved dependencies.
- (6/50) Late-evening and overnight lows are present, but no meal-linked carbohydrate or bolus data test this CR.
- (6/50) Late-evening and overnight lows create a safety constraint, but no meal-linked data establish that this CR is responsible.
- (5/50) Overnight lows could involve dinner insulin, but no dinner carbohydrate or bolus data exist.
- (3/50) Evening and overnight low exposure argues against adding meal insulin without meal-linked evidence.
- (3/50) Overnight lows may include residual evening meal or correction insulin, but no meal-linked evidence tests this CR.
- ... 10 more varying points

Points made by a single run only: 37.

Phrases used by at least half the runs: “overnight lows” (29), “meal linked” (29), “evening and overnight” (29).

Evidence cited: 19:00 (1), 00:00 (1).

decision: Profile / ISF 00:00

keep — 50/50 runs

Varying reasoning

- (15/50) Correction responses vary widely, while Dynamic ISF produced effective sensitivity from approximately 27.8–124 mg/dL/U.
- (9/50) Correction responses are highly variable and confounded, while Dynamic ISF produced effective sensitivity values far from the fixed ISF.
- (6/50) Correction windows include large delayed declines, but Dynamic ISF changes effective sensitivity to approximately 27.8–124 mg/dL/U and corrections are not isolated from SMBs, temporary basal, prior IOB, food, or activity.
- (6/50) The profile ISF cannot be isolated from SMB/UAM, temporary basal, meals, and IOB.
- (6/50) Fixed ISF should remain unchanged until automation and overlapping IOB are separated.
- (5/50) The base ISF cannot be isolated and should not be changed together with Dynamic ISF or SMB.
- ... 12 more varying points

Points made by a single run only: 26.

Phrases used by at least half the runs: “dynamic isf” (50), “effective sensitivity” (30), “static isf” (28), “isf cannot” (25).

Evidence cited: 124mg/dl (19), 56mg/dl (3), 43mg/dl (1).

decision: Profile / target 00:00

keep — 32/50 runs

Varying reasoning

- (6/32) A fasting-like overnight comparison is required before a numerical target change is supported.
- (5/32) Overnight low exposure makes this target relevant, but actual dosing also depends on residual IOB, temporary targets, SMB/UAM, Dynamic ISF, and basal.
- (3/32) A 90 mg/dL overnight target may contribute to insulin pressure, but residual IOB and automation are unresolved.
- (2/32) The target may provide limited margin during overnight IOB-associated declines, but the data do not separate target effects from basal, late insulin, and low treatment.
- (2/32) TBR is elevated from 00:00–03:00, making this a priority for verification.
- (2/32) This target overlaps substantial overnight low exposure and may provide limited margin, but residual evening IOB is an unresolved competing cause.
- ... 6 more varying points

Points made by a single run only: 35.

Evidence cited: 90mg/dl (4), 00:00 (2), 03:00 (2), 23:00 (1), 08:00 (1), 04:00 (1).

change — 18/50 runs

Shared reasoning

- (18/18) The 00:00 target overlaps recurrent 00:00 to 03:00 hypoglycemia, including 18.5% below range at 01:00.

Varying reasoning

- (4/18) Evening IOB remains uncertain, so this should be the only initial change.
- (3/18) Although prior IOB and automation remain uncertain, raising the target is a conservative single-setting candidate that does not assume a basal defect.
- (2/18) Corrections, meals, SMB/UAM, and Dynamic ISF remain uncertain dependencies, so this target candidate should be evaluated alone rather than accompanied by insulin-delivery changes.
- (2/18) Raising this target is a conservative safety-oriented candidate that reduces insulin pressure, although evening IOB remains uncertain.
- (2/18) A modest increase to 100 mg/dL is a safety-oriented candidate that avoids changing basal or automation.

Points made by a single run only: 15.

Evidence cited: 03:00 (17), 00:00 (16), 18.5% (13), 100mg/dl (7), 01:00 (6), 10.1% (3), 11.4% (3), 02:00 (3), 10% (3), 90mg/dl (1), 04:00 (1), 08:00 (1).

decision: Profile / target 08:00

keep — 50/50 runs

Varying reasoning

- (11/50) Daytime highs coexist with later lows, so lowering this target could increase insulin accumulation.
- (8/50) Meal execution, Dynamic ISF, SMB/UAM, corrections, and IOB remain unresolved dependencies, so changing this target is unsupported.
- (6/50) Daytime highs are meal- and automation-confounded, while later lows argue against a lower target.
- (5/50) Daytime highs and subsequent lows do not establish that this target is too high or too low.
- (4/50) Daytime highs are confounded by unrecorded meals, UAM, Dynamic ISF, and bolus timing.
- (4/50) Lowering the target could increase IOB and later hypoglycemia, while raising it cannot be justified without separating meal effects.
- ... 13 more varying points

Points made by a single run only: 36.

Phrases used by at least half the runs: “daytime highs” (45), “later lows” (27).

Evidence cited: 100mg/dl (1).

decision: Profile / target 23:00

keep — 48/50 runs

Varying reasoning

- (8/48) Keep this segment during the first 00:00 target test; later review depends on 23:00–00:00 glucose and preceding IOB.
- (7/48) Low exposure is elevated at 23:00, but changing both adjacent target segments together would obscure attribution.
- (5/48) Low exposure increases around 23:00-02:00, but the target change coincides with uncertain evening IOB and automated delivery.
- (5/48) Evidence: late-evening and overnight time below range is elevated after this segment begins.
- (4/48) Below-range exposure is elevated at 23:00 and afterward, but the effect of the target cannot be separated from evening correction and automated IOB.
- (3/48) Low exposure begins around 23:00, but this one-hour segment is confounded by evening IOB and the subsequent 00:00 target.
- ... 11 more varying points

Points made by a single run only: 43.

Evidence cited: 23:00 (39), 00:00 (18), 03:00 (5), 10.1% (4), 19:00 (3), 90mg/dl (1), 04:00 (1), 02:00 (1).

change — 2/50 runs

Shared reasoning

- (2/2) Evening IOB is an unresolved dependency, but changing this adjacent row with the 00:00 row creates one coherent 100 mg/dL overnight target.
- (2/2) Local 23:00 TBR is 10.1% and the low pattern continues after midnight.

Phrases used by at least half the runs: “overnight target” (2), “00 00 row” (2), “tbr at 23” (1), “100 mg/dl overnight” (1), “changing this adjacent” (1), “creates one coherent” (1), “falling glucose pattern” (1), “peak evening iob” (1), “00 and falling” (1), “one coherent 100” (1), “low risk peak” (1), “overnight low risk” (1).

Evidence cited: 23:00 (2), 10.1% (2), 00:00 (2), 100mg/dl (1).

decision: Profile / DIA

keep — 50/50 runs

Varying reasoning

- (6/50) Evidence: several response windows continue declining over two to six hours.
- (4/50) DIA depends on the insulin model, complete delivery history, Dynamic ISF, and SMB/UAM, so changing it could conceal another cause.
- (4/50) Isolated action curves are absent, and shortening DIA could undercount residual insulin.
- (3/50) Several excursions continue falling four to six hours after correction activity, supporting continued recognition of prolonged IOB.
- (3/50) Glucose responses continue through four to six hours, and substantial residual IOB is a safety concern.
- (3/50) Changing DIA would alter IOB and automation interpretation and should not be combined with ISF or SMB changes.
- ... 14 more varying points

Points made by a single run only: 53.

Phrases used by at least half the runs: “six hours” (28).

Evidence cited: 10h (6).

decision: AAPS / SMB and UAM

keep — 50/50 runs

Varying reasoning

- (7/50) Evidence: UAM and SMB manage rises when COB is zero and may contribute to accumulated IOB.
- (6/50) SMB/UAM respond to unannounced meal-like rises and may contribute to accumulated IOB, but Dynamic ISF, corrections, targets, and missing meal data are unresolved dependencies.
- (4/50) Meal execution, Dynamic ISF, corrections, DIA, and IOB remain unresolved dependencies.
- (4/50) SMB/UAM responds to unexplained rises while COB is zero, but UAM is not evidence of duplicate COB counting.
- (3/50) SMB/UAM respond to observed rises with COB at zero and may contribute to later IOB, but the evidence does not isolate their net effect from meals, Dynamic ISF, target, CR, and corrections.
- (3/50) SMB/UAM plausibly contributes to reactive insulin accumulation, but it is also compensating for unannounced rises.

- ... 10 more varying points

Points made by a single run only: 60.

Phrases used by at least half the runs: “dynamic isf” (32).

decision: AAPS / SMB activation conditions

keep — 50/50 runs

Varying reasoning

- (7/50) The activation settings are internally consistent, and UAM is not evidence of duplicate COB counting.
- (6/50) Always-enabled SMB creates broad eligibility, while disabling SMB with a high temporary target is safety-oriented.
- (4/50) Its contribution to lows cannot be separated from SMB limits, Dynamic ISF, meals, and IOB, so a change is unsupported.
- (4/50) Broad SMB availability may contribute to IOB, but no activation condition is isolated as causal and meal execution is unknown.
- (4/50) Changing activation together with UAM, meal announcements or Dynamic ISF would obscure attribution.
- (3/50) Broad SMB availability is a plausible contributor to IOB accumulation, but no episode isolates an activation condition from UAM, Dynamic ISF, corrections, targets, and meals.
- ... 9 more varying points

Points made by a single run only: 61.

Phrases used by at least half the runs: “dynamic isf” (33).

Evidence cited: 5.1% (1).

decision: AAPS / SMB strength and frequency

keep — 50/50 runs

Varying reasoning

- (4/50) Dynamic ISF, corrections, meals, temporary basal, and DIA remain dependencies, so no supported alternative interval or limit is available.
- (4/50) Frequent opportunities and the UAM limit could contribute to cumulative IOB, but the package lacks complete SMB sequences and dose totals.
- (3/50) The larger UAM limit is a plausible contributor to IOB accumulation, but representative records do not quantify the complete SMB sequence or isolate it from Dynamic ISF and temporary basal.
- (3/50) Frequent opportunities and the higher UAM allowance could accumulate insulin, but complete SMB sequences are unavailable and these settings may compensate for unannounced meals.
- (3/50) Keep these values during the target assessment; later review depends on complete high-to-low event traces.
- (3/50) No replacement interval or basal-minute limit is supported.
- ... 13 more varying points

Points made by a single run only: 58.

Phrases used by at least half the runs: “dynamic isf” (39).

decision: AAPS / max basal and max IOB limits

keep — 50/50 runs

Varying reasoning

- (10/50) These limits are permissive relative to scheduled basal and observed IOB, creating a material review concern.
- (8/50) The limits are permissive relative to scheduled basal and observed IOB, but neither ceiling was reached and no exact safer replacement is supported.
- (4/50) The limits are permissive relative to scheduled basal, representative IOB, and average daily insulin, creating a safety-review concern.
- (2/50) The ceilings are permissive relative to the profile and deserve safety review, but observed IOB did not approach 25 U and lower effective maxSafeBasal constraints appeared in loop records.
- (2/50) These ceilings are permissive relative to observed insulin use and deserve priority review, but they are not equivalent to delivered doses and lower derived caps were active in representative records.
- (2/50) The absolute limits are permissive relative to profile basal and observed IOB, but representative records show lower effective basal constraints and do not identify a safe replacement value.
- ... 8 more varying points

Points made by a single run only: 64.

Phrases used by at least half the runs: “permissive relative” (40).

Evidence cited: 25u (8), 4u/h (4), 5.935u (3), 6u (3), 0.8u/h (3), 5.94u (1), 12u/h (1).

decision: AAPS / Autosens and Dynamic ISF

keep — 50/50 runs

Varying reasoning

- (9/50) A sensitivity ratio of 1 is neutral and does not contradict disabled Autosens.
- (7/50) Dynamic ISF is a plausible contributor because variable sensitivity became substantially stronger than the static ISF during highs.
- (7/50) Evidence: Dynamic ISF materially strengthens effective sensitivity during highs, while Autosens is disabled and a sensitivity ratio of one is neutral.
- (6/50) Dynamic ISF materially strengthens correction during some highs and may contribute to late lows, but its effect cannot be separated from UAM, SMB, meals, static ISF, and IOB.
- (5/50) Dynamic ISF is the leading automation hypothesis because effective sensitivity strengthened markedly during highs, but no supported replacement percentage is available.
- (3/50) Dynamic ISF sometimes makes effective sensitivity much stronger during highs and may contribute to accumulated IOB, but meal effects and SMB/UAM are unresolved.
- ... 14 more varying points

Points made by a single run only: 48.

Phrases used by at least half the runs: “dynamic isf” (49).

Evidence cited: 56mg/dl (1), 42.5mg/dl (1), 70% (1).

decision: AAPS / carbohydrate absorption model

keep — 50/50 runs

Varying reasoning

- (13/50) No regular carbohydrate or e-carb entries exist, so the absorption settings cannot be compared with an announced meal.
- (11/50) Prolonged rises cannot be attributed to absorption without meal-linked evidence separating IOB, SMB, corrections, and temporary targets.
- (9/50) Meal-linked amount, timing, absorption, IOB, SMB, temporary-target, and correction evidence is required before changing either component.
- (7/50) There are no regular-carb or e-carb entries against which to evaluate absorption.
- (5/50) No carbohydrate or e-carb entries test these settings, and COB is zero in representative loops.
- (4/50) No carbohydrate or e-carb entry exercised the COB absorption model during this period.

- ... 10 more varying points

Points made by a single run only: 31.

Phrases used by at least half the runs: “meal linked” (40), “carb entries” (32), “carbohydrate or carb” (27).

profile recommendation

target — 39/50 runs

Varying reasoning

- (16/39) Local time below range was 11.4% at 00:00, 18.5% at 01:00, 10.1% at 02:00, and 10.2% at 03:00.
- (13/39) Keep the current profile, including the 90 mg/dL overnight target, until an overnight low-confounder review separates target and basal effects from residual IOB.
- (10/39) Review changing the 00:00 target from 90 mg/dL to 100 mg/dL as the only initial profile change.
- (8/39) The overnight low pattern is strong, but the package cannot separate the target or 00:00 basal from residual evening IOB, SMB/UAM, Dynamic ISF, corrections, activity, and low treatment.
- (8/39) Review changing the 00:00 target from 90 to 100 mg/dL while keeping all basal, CR, ISF, DIA, and AAPS settings unchanged during the evaluation.
- (6/39) No profile row has sufficiently isolated evidence for a supported old-to-new value.
- ... 14 more varying points

Points made by a single run only: 39.

Phrases used by at least half the runs: “90 mg/dl” (30), “100 mg/dl” (25).

Evidence cited: 90mg/dl (30), 100mg/dl (25), 03:00 (23), 00:00 (23), 18.5% (18), 23:00 (13), 01:00 (13), 02:00 (13), 08:00 (12), 11.4% (10), 04:00 (9), 10.1% (9).

basal — 11/50 runs

Varying reasoning

- (5/11) Keep the current profile and first verify the 19:00, 00:00, and 23:00 target segments using one fasting-like 23:00–03:00 interval with complete IOB and insulin context.
- (5/11) The recurrent overnight hypoglycemia is important, but the package cannot separate scheduled basal or the 90 mg/dL target from residual evening IOB, SMB/UAM, Dynamic ISF, corrections, and temporary basal.
- (4/11) Daytime highs are also meal-confounded, so they do not support strengthening basal, CR, or ISF.
- (3/11) The stable 04:00 to 08:00 interval argues against a broad basal increase or reduction.
- (2/11) Low exposure is recurrent and clinically important, but the supplied data cannot isolate basal from corrections, UAM/SMB, Dynamic ISF, meals, activity, temporary targets, rescue carbohydrates, or persistent IOB.
- (2/11) Keep all current profile segments unchanged and verify overnight basal and target behavior using at least three fasting-like, low-IOB overnight traces.

Points made by a single run only: 18.

Evidence cited: 00:00 (5), 03:00 (4), 90mg/dl (4), 23:00 (3), 08:00 (3), 04:00 (3), 19:00 (2), 18:00 (1), 06:00 (1).

primary recommendation

target — 28/50 runs

Varying reasoning

- (16/28) The recurrent 00:00-03:00 low pattern supports reviewing the 00:00 target before changing basal or automation.
- (8/28) Keep the profile unchanged while verifying the overnight low pattern.
- (5/28) The 23:00-03:00 low pattern is important, but the current package cannot separate the 90 mg/dL target and scheduled basal from evening IOB, SMBs, corrections, activity, or low treatment.
- (5/28) A candidate review from 90 to 100 mg/dL at 00:00 is more directly supported than changing basal, CR, ISF, DIA, or automation.
- (5/28) Primary: review the midnight target.
- (2/28) Review one continuous overnight target change first.
- ... 4 more varying points

Points made by a single run only: 29.

Phrases used by at least half the runs: “low pattern” (15).

Evidence cited: 03:00 (17), 00:00 (16), 100mg/dl (12), 90mg/dl (10), 23:00 (8), 04:00 (1), 10% (1), 19% (1).

basal — 9/50 runs

Varying reasoning

- (5/9) Keep the profile while verifying the overnight low pattern.
- (3/9) This is the most direct way to distinguish the 00:00 basal and 90 mg/dL target from residual evening insulin.
- (3/9) Keep the profile unchanged while verifying the low patterns.
- (2/9) Keep the current basal and target schedule while reviewing one fasting-like 23:00–03:00 interval with complete insulin, IOB, temporary-target, food, correction, and activity context.

Points made by a single run only: 14.

Phrases used by at least half the runs: “keep the profile” (7), “overnight low” (6), “temporary target” (5).

Evidence cited: 23:00 (2), 03:00 (2), 90mg/dl (2), 00:00 (1).

profile — 8/50 runs

Varying reasoning

- (4/8) Keep the profile unchanged while verifying the recurrent low pattern.
- (2/8) This is the clearest way to separate the 00:00 basal and 90 mg/dL target from residual evening insulin.
- (2/8) No basal, target, CR, ISF, or DIA segment has enough isolated evidence for a numeric change.
- (2/8) Primary: keep the profile stable and verify the low pattern first.

Points made by a single run only: 14.

Phrases used by at least half the runs: “low pattern” (6), “basal target cr” (6), “target cr isf” (6), “isf or dia” (4), “keep the profile” (4).

Evidence cited: 23:00 (3), 03:00 (2), 00:00 (2), 90mg/dl (2), 18:00 (1), 19:00 (1), 04:00 (1).

safety_and_data — 2/50 runs

Shared reasoning

- (2/2) Each timeline should distinguish glucose, total insulin, IOB, SMB, temp basal, corrections, temporary targets, meals, activity, and low treatment.

Points made by a single run only: 4.

Phrases used by at least half the runs: “profile and automation” (2), “activity and low” (2), “low treatment” (2), “temporary targets” (2), “low glucose sequences” (1), “corrections temporary targets” (1), “insulin iob smb” (1), “basal corrections temporary” (1), “recurrent low glucose” (1), “timeline should distinguish” (1), “evening low patterns” (1), “midnight to 03” (1).

Evidence cited: 03:00 (1).

data_and_profile — 2/50 runs

Points made by a single run only: 6.

Phrases used by at least half the runs: “low pattern” (2), “changing the profile” (1), “low iob meal” (1), “primary next review” (1), “complete glucose” (1), “verify the overnight” (1), “insulin delivery context” (1), “review is one” (1), “pattern before changing” (1), “one low iob” (1), “glucose and insulin” (1), “90 mg/dl target” (1).

Evidence cited: 00:00 (1), 90mg/dl (1).

safety — 1/50 runs

Points made by a single run only: 3.

Phrases used by at least half the runs: “bolus smb temporary” (1), “overnight lows” (1), “reconstruct severe low” (1), “severe low insulin” (1), “temporary basal target” (1), “smb temporary basal” (1), “target effects” (1), “primary reconstruct severe” (1), “reviewing complete glucose” (1), “separates residual insulin” (1), “profile and automation” (1), “low insulin sequences” (1).

Evidence cited: 19:00 (1).

summary

50 runs

Shared reasoning

- (50/50) Fourteen days of high-quality CGM data show strong overall time in range of 86.0%, but time below range of 5.1% is the main safety concern.

Varying reasoning

- (14/50) Daytime rises, especially from 14:00 to 17:00 local time, frequently occur with no recorded carbohydrates and are managed through UAM, SMB, temporary basal, corrections, and Dynamic ISF.
- (13/50) The primary conservative profile review is a candidate increase of the 00:00 target from 90 to 100 mg/dL, with no simultaneous basal or automation changes.
- (12/50) Missing meal and carbohydrate records prevent separation of meal timing, CR, delayed absorption, correction stacking, SMB/UAM, Dynamic ISF, activity, and rescue-carbohydrate effects.
- (10/50) The safest synthesis is to keep the current profile and AAPS values while first obtaining an interpretable overnight observation and then one fully documented meal response.
- (7/50) No carbohydrate or meal-strategy entries were declared or recorded, so basal, carbohydrate ratio, ISF, DIA, meal timing, and absorption effects cannot be separated reliably.
- (6/50) No profile or AAPS setting has sufficiently isolated evidence for a numerical change.
- ... 20 more varying points

Points made by a single run only: 33.

Phrases used by at least half the runs: “time in range” (50), “fourteen days” (49), “high quality cgm” (49), “days of high” (49), “dynamic isf” (45), “strong overall” (40), “below range” (30), “strong overall time” (29), “time below range” (25).

Evidence cited: 86.0% (30), 5.1% (29), 18:00 (22), 19:00 (22), 03:00 (21), 00:00 (16), 42mg/dl (16), 121.6mg/dl (15), 100mg/dl (14), 17:00 (12), 23:00 (10), 14:00 (9).

issues

50 runs

Shared reasoning

- (50/50) Time below range was 5.1%, with recurrent severe readings of 41–42 mg/dL and concentrated low exposure around 23:00–03:00 and 18:00–19:00 local time.
- (35/50) No carb or e-carb entries and no declared meal-strategy values were available, preventing reliable CR, prebolus, immediate bolus percentage, or absorption assessment.

Varying reasoning

- (23/50) The 12 U/h maximum basal and 25 U maximum IOB are permissive relative to scheduled basal, although the telemetry does not show that either hard ceiling directly caused an event.
- (23/50) Dynamic ISF sometimes produced substantially stronger effective sensitivity than the static ISF while SMB/UAM and temporary basal were also active.
- (16/50) Absolute max basal and max IOB limits are permissive relative to observed delivery, although the data do not show either ceiling being reached.
- (10/50) No carbohydrate treatments or user-declared meal strategy were available, preventing separation of meal timing, CR, absorption, SMB/UAM, and correction effects.
- (9/50) Afternoon highs are sometimes followed by evening lows, suggesting a high-to-low cycle involving corrections, Dynamic ISF, SMB/UAM, temporary basal, and residual IOB.
- (9/50) Daytime rises are sometimes followed by substantial evening or delayed declines, suggesting insulin accumulation rather than a simple need for stronger basal or meal dosing.
- ... 17 more varying points

Points made by a single run only: 33.

Phrases used by at least half the runs: “dynamic isf” (49), “time below range” (44), “42 mg/dl” (41), “meal strategy” (40), “low exposure” (33), “41 42 mg/dl” (31), “range is 5.1” (30), “cr prebolus” (29), “immediate bolus” (29), “declared meal” (28), “declared meal strategy” (27), “immediate bolus percentage” (26).

Evidence cited: 5.1% (50), 42mg/dl (41), 19:00 (30), 18:00 (28), 03:00 (22), 25u (14), 12u/h (12), 00:00 (11), 23:00 (10), 56mg/dl (8), 90mg/dl (4), 17:00 (3).

step findings: data_quality

50 runs

Shared reasoning

- (33/50) CGM coverage and cadence support time-of-day and safety analysis.

Varying reasoning

- (27/50) The authoritative AAPS profile is complete and declared historically applicable to the period.
- (23/50) Meal-specific analysis remains limited because the declared meal strategy is empty and no carbohydrate treatments were recorded.
- (8/50) CGM and profile data are sufficiently complete for time-of-day and safety analysis.
- (8/50) The authoritative top-level AAPS profile is complete, and profile_context declares it historically applicable.
- (5/50) Event-level causal attribution is weaker because meal, carbohydrate, activity, and rescue-carbohydrate records are absent.
- (3/50) CGM and general Nightscout coverage are sufficient for identifying recurring clock-time patterns.
- ... 8 more varying points

Points made by a single run only: 11.

Phrases used by at least half the runs: “profile is complete” (36), “cgm coverage” (36), “time of day” (36).

step findings: safety_overview

50 runs

Shared reasoning

- (30/50) Hypoglycemia is the dominant safety constraint despite strong overall time in range.

Varying reasoning

- (8/50) Low exposure is concentrated overnight and around 18:00-19:00 local time.
- (8/50) Recurrent overnight lows and daytime high-to-low transitions limit any proposal to strengthen insulin delivery.
- (8/50) Recurrent hypoglycemia is the principal safety constraint.
- (5/50) Excess hypoglycemia is the main safety constraint.
- (5/50) Hypoglycemia is the primary safety constraint.
- (3/50) Afternoon highs followed by evening lows suggest delayed cumulative insulin exposure rather than a simple need for stronger profile dosing.
- ... 11 more varying points

Points made by a single run only: 30.

Phrases used by at least half the runs: “safety constraint” (43).

Evidence cited: 18:00 (7), 19:00 (7), 03:00 (3), 42mg/dl (2), 00:00 (2), 23:00 (1), 40mg/dl (1).

step findings: basal_targets

50 runs

Varying reasoning

- (22/50) The recurring 00:00–03:00 low pattern supports reviewing the 00:00 target before changing basal.
- (21/50) The 04:00 to 08:00 interval is comparatively stable.
- (9/50) The transition from afternoon highs to 18:00-19:00 lows argues against increasing afternoon or evening basal from these aggregate data.
- (6/50) Daytime highs cannot be assigned to insufficient basal because meals and corrections remain plausible.
- (6/50) A recurrent overnight low pattern exists, but scheduled basal and the 90 mg/dL overnight target cannot be separated from evening IOB and automation.
- (5/50) The 04:00-07:00 period is comparatively stable and does not identify the 05:00 or 06:00 basal segments as recurring defects.
- ... 20 more varying points

Points made by a single run only: 36.

Phrases used by at least half the runs: “daytime highs” (27).

Evidence cited: 08:00 (25), 03:00 (24), 90mg/dl (24), 00:00 (18), 04:00 (18), 19:00 (12), 18:00 (11), 23:00 (10), 06:00 (9), 05:00 (7), 16:00 (5), 17:00 (3).

step findings: profile_cr_isf_dia

50 runs

Shared reasoning

- (31/50) The static ISF cannot be isolated because Dynamic ISF materially changes effective sensitivity.

Varying reasoning

- (20/50) No CR segment can be tested without meal-linked carbohydrate and bolus data.
- (12/50) No CR segment can be validated without known carbohydrate amounts and meal timing.
- (6/50) Delayed declines and persistent IOB do not support shortening the 10-hour DIA.
- (6/50) Static ISF cannot be separated from Dynamic ISF, SMB, temporary basal, and other IOB.
- (5/50) The 10-hour DIA may help preserve recognition of prolonged insulin action, and shortening it would be unsupported.
- (4/50) The 10-hour DIA cannot be validated from mixed six-hour response windows.
- ... 14 more varying points

Points made by a single run only: 42.

Phrases used by at least half the runs: “dynamic isf” (46), “hour dia” (41), “10 hour dia” (40), “cr segment” (34), “static isf” (26), “isf cannot” (25).

Evidence cited: 56mg/dl (13), 10h (5), 16:00 (2), 124mg/dl (1), 10.5g/u (1).

step findings: automation_smb_uam

50 runs

Varying reasoning

- (11/50) UAM is not duplicate COB counting.
- (8/50) SMB and UAM are internally consistent and respond to unannounced glucose rises.
- (8/50) SMB and UAM are active and appear to respond to rises while COB is zero.
- (5/50) UAM is not duplicate COB accounting.
- (4/50) Carbohydrate absorption settings cannot be evaluated in observed use because no carbohydrate or e-carb entries exist.
- (3/50) Broad SMB eligibility and the larger UAM SMB envelope may contribute to cumulative IOB, but no individual SMB setting is proven causal.
- ... 32 more varying points

Points made by a single run only: 47.

Phrases used by at least half the runs: “smb and uam” (33).

Evidence cited: 15min (1).

step findings: automation_sensitivity_limits

50 runs

Varying reasoning

- (21/50) The configured max basal and max IOB limits are permissive relative to scheduled basal and observed IOB, but telemetry does not show either configured ceiling being reached.
- (15/50) The 12 U/h maximum basal and 25 U maximum IOB are permissive backstops relative to the profile, but the package does not support safe replacement values.
- (9/50) Dynamic ISF is a plausible contributor to strong high-glucose responses because effective sensitivity became much stronger than the static ISF during some highs.
- (7/50) Dynamic ISF can make effective sensitivity substantially stronger during highs.
- (7/50) A sensitivity ratio of 1 in loop telemetry represents a neutral current adjustment and does not contradict disabled Autosens.
- (7/50) Carbohydrate absorption settings cannot be evaluated because there are no announced carbohydrate traces.
- ... 16 more varying points

Points made by a single run only: 44.

Phrases used by at least half the runs: “dynamic isf” (50), “maximum basal” (25), “max iob” (25), “max basal” (25).

Evidence cited: 12u/h (16), 25u (13), 70% (2), 80mg/dl (1), 4u/h (1), 56mg/dl (1).

step findings: meal_bolus_strategy

50 runs

Varying reasoning

- (21/50) No numerical prebolus interval or immediate bolus percentage can be inferred.
- (19/50) The user-declared analysis context contains no meal strategy, so no declared prebolus interval or immediate bolus percentage exists.
- (8/50) Daytime rises could reflect timing, CR, delayed absorption or unannounced meals, while later lows make additional immediate or corrective insulin unsafe to infer from glucose alone.
- (7/50) No typical meal strategy was declared in aaps_profile.analysis_context, and observed telemetry contains no meal carbohydrate entries.
- (6/50) There is no declared numeric prebolus or immediate meal-bolus strategy, and observed execution cannot be reconstructed.
- (4/50) The recurring daytime rise followed by evening decline is compatible with reactive UAM/SMB coverage and delayed IOB, but it does not support a stronger CR, earlier prebolus, or larger immediate bolus percentage.
- ... 13 more varying points

Points made by a single run only: 51.

Phrases used by at least half the runs: “immediate bolus” (47), “immediate bolus percentage” (45), “meal strategy” (25).

Evidence cited: 16:00 (1).

step findings: meal_ecarbs_absorption

50 runs

Shared reasoning

- (34/50) No e-carb amount, start offset or duration can be inferred.

Varying reasoning

- (16/50) E-carbs are not equivalent to a pump extended bolus.
- (6/50) No meal-linked evidence supports introducing e-carbs or changing the carbohydrate absorption model.
- (5/50) Prolonged rises cannot be attributed to delayed absorption without separating meal, IOB, SMB, temporary-target, correction, and activity effects.
- (5/50) E-carbs cannot be inferred from correction-response windows and are not equivalent to a pump extended bolus.
- (5/50) No regular-carb or e-carb absorption strategy can be evaluated.
- (4/50) Prolonged highs cannot be attributed to delayed absorption without meal-linked evidence, and e-carbs must not be treated as an extended pump bolus.
- ... 11 more varying points

Points made by a single run only: 27.

Phrases used by at least half the runs: “carb amount” (31), “extended bolus” (30), “pump extended bolus” (28), “offset or duration” (27).

step findings: synthesis_plan

50 runs

Varying reasoning

- (15/50) The primary candidate is changing only the 00:00 target from 90 to 100 mg/dL.
- (7/50) No profile or AAPS value has sufficiently isolated evidence for a supported numeric change.
- (5/50) Basal, CR, ISF, DIA, SMB/UAM, Dynamic ISF, safety limits, and carb absorption should otherwise remain unchanged until the target adjustment is observed and better contextual data are collected.
- (4/50) The leading explanation for daytime high-to-evening-low sequences is interaction among undocumented meal-like rises, UAM/SMB catch-up, Dynamic ISF, corrections, and accumulated IOB.
- (4/50) The safest action order is to verify data and profile behavior first, review automation exposure second, and assess meal strategy last using a documented meal.
- (4/50) The safe order is profile and overnight verification first, safety-limit and automation review second, and documented meal-strategy testing third.
- ... 21 more varying points

Points made by a single run only: 82.

Evidence cited: 00:00 (16), 100mg/dl (15), 90mg/dl (6), 23:00 (3), 03:00 (2).

meal strategy

50 runs

Varying reasoning

- (22/50) The declared meal-strategy list is empty and Nightscout contains no carb or e-carb entries.
- (20/50) Keep SMB/UAM unchanged during structured observation, and do not introduce an e-carb amount, offset, or duration without a recurring meal-linked delayed-absorption pattern.
- (20/50) Do not assign a numerical prebolus interval or immediate bolus percentage from this package.
- (7/50) No meal strategy is declared and no carbohydrate treatments are recorded.
- (6/50) Keep prebolus timing and immediate bolus percentage unspecified, keep SMB/UAM settings unchanged during the initial review, and do not introduce e-carbs without meal-linked evidence.
- (6/50) Do not assign a numeric prebolus or immediate bolus percentage from this package.
- ... 20 more varying points

Points made by a single run only: 66.

Phrases used by at least half the runs: “immediate bolus percentage” (50), “meal strategy” (44), “dynamic isf” (39), “meal linked” (37), “daytime rises” (25), “declared meal strategy” (25).

Evidence cited: 5.1% (1).